

CARLOS THERAN-SUAREZ, PhD

Tallahassee, FL, 32301 | carlos.theran@fam.u.edu

ASSISTANT PROFESSOR

Computer Vision — AI Decision Making — Predictive Analysis — Natural Language Processing

Experienced data scientist with 10+ years specializing in developing, implementing, and executing machine learning and deep learning pipelines for computer vision, natural language processing, and healthcare applications, resulting in improving literature results and documented as scientific publications. Excelled in collaborative, interdisciplinary environments, contributing to impactful research and problem-solving, and showcasing strong communication skills through numerous presentations, and workshops. My expertise lies in algorithm design, machine learning engineering, statistical analysis, numerical analysis, and strategic optimization. Skilled in analyzing data to drive decision-making and succeed in fast-paced environments where creativity and innovation are valued. Proficient in TensorFlow, Python, Scikit-learn, and OpenCV.

Advanced Machine Learning / Deep Learning techniques | Data Modeling | Visualization | Problem Resolution | Containerization Deployment | Generative AI | Cloud ML Services | Technical and Scientific Writing.

TECHNICAL SKILLS

Programming languages and Software: Python, C++, Java, MATLAB, Hadoop, Spark, Hive, HTML, CSS, Docker, Kubernetes, SQL.

Machine Learning Libraries: Pandas, Matplotlib, NumPy, scikit-learn, Seaborn, Keras, TensorFlow, Transformers.

Cloud Services: AWS SageMaker, Google Cloud, Microsoft Azure, National Research Platform (Nautilus).

Collaborative Coding Software: GitHub, GitLab, and Docker Hub.

PROFESSIONAL EXPERIENCE

FLORIDA A&M UNIVERSITY, Tallahassee, FL

FGLSAMP Mentor

08/2023 – Present

Since August 2023, I have mentored approximately 30 undergraduate students in interdisciplinary research projects spanning healthcare, remote sensing and geospatial analysis, and AI-driven image analysis. My role has involved guiding students through the research process, from problem formulation and methodology development to implementation, analysis, and dissemination of results. These mentorship activities have provided students with meaningful exposure to applied machine learning and deep learning research while strengthening their preparation for graduate study and professional careers. As a result of this mentoring, students have successfully presented their work at scholarly venues including the Florida Undergraduate Research Conference, the Undergraduate Research Symposium at Florida A&M University, and the McKnight Doctoral Fellowship Annual Fellows Meeting. This mentorship has also contributed to important student outcomes, including one mentee being accepted into a post-baccalaureate program at Moffitt Cancer Center and others obtaining opportunities to work in industry. Research topics includes. Predicting diabetic patients using machine learning models, Machine learning framework for lung cancer detection using histopathological images, Land crop monitoring using machine learning models, Evaluating cooling methodologies for modern AI data centers, YOLO-based crowd anomaly and threat detection, Non-invasive Alzheimer's detection via MRI imaging and convolutional neural networks

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Researcher

01/2021 – Present

Lead role in developing and implementing machine learning and deep learning pipelines specifically tailored for the healthcare sector. These pipelines leverage advanced algorithms and data analytics techniques to enhance medical diagnostics, patient care, and treatment outcomes. I specialize in teaching diverse range of programming courses that equip students with essential skills in various domains. These courses include programming languages (Java, Python, and C++), network-based multimedia (HTML, CSS), object-oriented programming (C++), statistics and probability (Python, Seaborn), Data Science (Python, Machine Learning Libraries), and cloud computing (Hadoop, Hive, Spark).

- Developed and validated machine learning models (such as Support Vector Machine, Neural Networks, XGBoost, and Decision Tree) to leverage healthcare data analysis in predicting diabetes stage in patients, demonstrating commitments to cutting-edge research methodologies. Conducted in collaboration with the College of Pharmacy at Florida A&M University; and implemented using Python, along with libraries such as Scikit-learn and TensorFlow.
- Designed and executed pipelines grounded in state-of-the-art machine learning models for precise categorization of Alzheimer's stages through MRI image information. Resulted in the publication of a peer-reviewed conference paper, which was recognized and awarded as the best paper in its category; and implemented using Python along with libraries such as Seaborn, Scikit-learn and TensorFlow.
- Provided mentorship to undergraduate and graduate students in research topics focused on delivering machine learning-based solutions for blockchain-based cryptocurrency, breast cancer detection, and facial recognition for autism detection. The successful outcomes of these efforts culminated in peer-reviewed journal publication; and implemented in Python along with libraries such as OpenCV, TensorFlow, Keras, Scikit-learn, and other machine learning libraries.
- Conducted assessment of pre-trained Large Language models (SciBERT and BERT) to enhance health communication, particularly in the context of doctor-patient interactions and addressing disparities. The findings of this research were successfully documented and published in a peer-reviewed paper conference.
- Facilitated deep understanding of programming concepts, algorithm design, data analysis, and cloud infrastructure management through comprehensive instruction and practical demonstrations.

Assistant Professor

01/2021 – Present

- As an Assistant Professor, I have contributed to student learning through the development and delivery of courses that combine rigorous theoretical instruction with applied, hands-on experiences in computer science, data science, and artificial intelligence. My teaching philosophy emphasizes active learning, critical thinking, and the integration of real-world problems into the classroom to strengthen students' technical competence and intellectual growth. I design lectures, assignments, and experiential activities that encourage students to connect foundational concepts with practical implementation, thereby enhancing their preparedness for research, graduate education, and industry opportunities. Through this instructional role, I seek to cultivate a learning environment that is both academically challenging and supportive of student development. Courses Taught; Design and Analysis of Algorithms, Programming Language, Database Management System, Object-Oriented Programming, Special Topic [Intro to GenAI to Build Machine Learning and Deep Learning models, Design and Analysis of Algorithms, Programming Language, Network-Based Multimedia, Special Topic [Intro to Data Sciences], Foundation of Computer Sciences, Intro to Artificial Intelligent.

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JOHN CABOT UNIVERSITY, Rome, Italy

Visiting Professor

summers; 2022, 2023, 2024, 2025, 2026

Teaching college-level courses in Machine Learning with Python, Probability and Statistics, and Fundamentals of Programming with Python.

- Designed course materials, delivered lectures, conducted practical sessions, assessing student performance, and provided academic support.
- Updated with industry trends, incorporating relevant case studies and real-world applications into the curriculum, and fostering a dynamic learning environment to help students develop strong foundational knowledge and practical skills in data science.

NSF-EPSCoR CENTER FOR THE ADVANCEMENT OF WEARABLE TECHNOLOGIES, Mayaguez, Puerto Rico

Research Assistant

01/2020 – 12/2020

Designing and architecting cloud-based data solutions, including selecting appropriate cloud platforms (e.g., Azure, AWS, Google Cloud), services, and tools based on project requirements. As well as developing data pipelines for ingesting, processing, and transforming large volumes of data from diverse sources into usable formats for analysis and storage.

- Built a cloud computing ecosystem using Hadoop, Kafka, and Apache Spark for large-scale data analysis and process automatization. As a result, the findings of this research were successfully documented and published.
- Implemented a strategy rooted in machine learning and deep learning techniques for precise chemical component analysis. The implemented model for this project was ElemNet.
- Designed and implemented a WebApp that seamlessly integrates cloud computing tools, establishing a Software as a Service (SaaS) platform for enhanced functionality and accessibility of the developed cloud computing ecosystem.

KEY ACADEMIC PROJECTS

UNIVERSITY OF PUERTO RICO, Mayagüez, Puerto Rico

Graduate Research Assistant

06/2017 – 01/2021

Designed and implemented machine learning and deep learning models based on convolutional neural networks (CNNs) and recurrent neural networks (RNNs) algorithms for computer vision problems. Also, developing pipelines to preprocess (normalization, data augmentation, and feature extraction) large-scale visual datasets to train, test, and validate machine learning models.

- Designed and implemented innovative deep learning pipeline for LIDAR image super-resolution. The implemented Encoder-Decoder model estimates the missing pixel information; implemented in MATLAB.
- Designed and implemented pipelines for hyperspectral and multispectral images fusion by scalable resolution patches to enhance the hyperspectral spatial resolution using Long Short-Term Memory and Convolutional Neural Network models. Resulted in successfully documentation and publication; implemented in MATLAB.
- Applied machine learning models to improve land cover classification accuracy by applying data fusion techniques. Resulted in successfully documentation and publication; implemented in MATLAB.
- Maintained and updated the Hyperspectral Image Analysis MATLAB Toolbox (HIAT) with unsupervised and supervised machine learning algorithms for image analysis (such as Long Short-Term Memory, Convolutional Neural Network, Artificial Neural Network, and Support Vector Machine models, Euclidean Distance).

UNIVERSITY OF PUERTO RICO, Mayagüez, Puerto Rico

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Graduate Research Assistant

01/2013 – 12/2015

Developing advanced numerical analysis techniques to accurately approximate numerical solutions for challenging ill-conditioned problems.

- Developed a preconditioner block technique based on Incomplete Cholesky Factorization for ill-conditioned problems generated by Discontinuous Galerkin Method, implemented in C++.
- Implemented finite element method to approximate numerical solution for the Wave Equation, implemented in MATLAB.

EDUCATION

UNIVERSITY OF PUERTO RICO, Mayagüez, Puerto Rico

PhD in computer and information science and engineering, 2023

Relevant coursework: Machine Learning (unsupervised and supervised models), Deep Learning, Cloud-native Services, Image Analysis, Design of Algorithms, Probability and statistic.

UNIVERSITY OF PUERTO RICO, Mayagüez, Puerto Rico

Master of Science in Scientific Computing, 2015

Relevant coursework: Numerical Analysis, Preconditioner Techniques, Mathematical Modeling, High Performance Computing, Partial Differential equation.

UNIVERSITY OF ATLÁNTICO, Barranquilla, Colombia

Bachelor of Science in Mathematical Science, 2011

Relevant coursework: Functional Analysis, Probability and Statistic, Analysis, High Performance Computing.

PROFESSIONAL CREDENTIALS

- Efficient Large Language Model (LLM) Customization, NVIDIA, March 2024.
- Generative AI with Large Language Models, Coursera/Deeplearning.ai, October 2023.
- Machine Learning Intensive Professional Development, The Coding School and AWS Machine Learning University, November 2023.
- Data Life-Cycle Training, Abstraction, Virtualization and Containerization for reproducible science, The University of North Carolina at Chapel Hill, June 2019.